# Evaluation runs: chain (s4-chain)

generated 2026-11-05 from brain-data a1150b5a2 2026-11-05 14:21:32 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): (the single pair): expected conforms (–threshold 65536 is needed at d¿=512). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{spec}$, $G_{impl}$ = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G^{expl}_{spec}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

### stateless: pair **single**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 1 | 1 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 8 | 1 | 1 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 32 | 1 | 1 | conforms | 0.0 | 4 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 128 | 1 | 1 | conforms | 0.1 | 14 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 1024 | 1 | 1 | conforms | 2.1 | 407 | 48 h, 250 GB | 1 | 1 | brain01, 2026-11-03 |
| 2048 | 1 | 1 | conforms | 9.4 | 1471 | 48 h, 250 GB | 1 | 1 | brain04, 2026-11-03 |
| 4096 | 1 | 1 | conforms | 35.5 | 5545 | 48 h, 250 GB | 1 | 1 | brain04, 2026-11-03 |
| 8192 | 1 | 1 | conforms | 160.3 | 21491 | 48 h, 250 GB | 1 | 1 | brain04, 2026-11-03 |

**Not run (stateless).** left out by the table rules (ok): 512 fifo.
the tool ended with an error (exit-101): 16384 fifo.
planned but never scheduled (grid points beyond the ladder): 32768 fifo.

## stateful

### stateful: pair **single**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2 | 1 | 1 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 8 | 1 | 1 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 32 | 1 | 1 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 128 | 1 | 1 | conforms | 0.1 | 12 | 1 h, 250 GB | brain02, 2026-11-02 |
| 1024 | 1 | 1 | conforms | 1.5 | 330 | 48 h, 250 GB | brain01, 2026-11-03 |
| 2048 | 1 | 1 | conforms | 5.3 | 1199 | 48 h, 250 GB | brain04, 2026-11-03 |
| 4096 | 1 | 1 | conforms | 21.1 | 4548 | 48 h, 250 GB | brain04, 2026-11-03 |
| 8192 | 1 | 1 | conforms | 108.9 | 14820 | 48 h, 250 GB | brain04, 2026-11-03 |
| 32768 | | | 🕓 (48 h) | | | 48 h, 250 GB | brain01, 2026-11-03 |

**Not run (stateful).** left out by the table rules (ok): 512 fifo, 16384 fifo.

## offline

### offline: pair **single**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 1 | 1 | conforms | 0.3 | 4 | 10 min, 16 GB | 0.3 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 8 | 1 | 1 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 32 | 1 | 1 | conforms | 0.0 | 4 | 1 h, 250 GB | 0.0 | 0.0 | 0.0 | merge | brain01, 2026-11-03 |
| 128 | 1 | 1 | conforms | 0.2 | 10 | 1 h, 250 GB | 0.2 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 512 | 1 | 1 | conforms | 0.5 | 80 | 48 h, 250 GB | 0.2 | 0.4 | 0.1 | merge | brain01, 2026-11-03 |
| 4096 | 1 | 1 | conforms | 28.5 | 3697 | 48 h, 250 GB | 7.1 | 22.8 | 5.7 | merge | brain01, 2026-11-03 |

**Not run (offline).** skipped: a smaller size had reached its limit: 1024 fifo, 16384 fifo, 32768 fifo.
the tool ended with an error (exit-127): 2048 fifo, 8192 fifo.